

AES-256 PDF Encryption

Zero-dependency password protection using the PDF Standard Security Handler, Revision 6 by default (legacy Revision 4 / AES-128 available). Implemented entirely with System.Security.Cryptography.

How It Works

By default, TerraPDF derives a 32-byte **File Encryption Key (FEK)** from your passwords using the PDF 2.0 SHA-2 based key-derivation algorithm (Algorithm 2.B, Revision 6). Each PDF object — pages, images, content streams, metadata, bookmark titles, hyperlink URIs — is then encrypted with a unique **per-object AES-256 CBC key** derived from the FEK plus the object number. Set `Algorithm = EncryptionAlgorithm.Aes128` to opt into the legacy 16-byte-key, MD5-derived Revision 4 handler instead (see 12e).

Component	Detail
Cipher	AES-256 CBC (per ISO 32000-2 §7.6.5)
IV	16 random bytes — unique per object
Key derivation	SHA-2 based, Algorithm 2.B (ISO 32000-2, Revision 6)
O/U entries	48-byte verifiers, with /OE + /UE key-wrapping
Handler	PDF Standard Security Handler, Revision 6
PDF version	2.0 (required for Revision 6)
Legacy mode	Algorithm = EncryptionAlgorithm.Aes128 ? AES-128 CBC, Revision 4, PDF 1.6
Dependencies	System.Security.Cryptography only — no packages

Five Protection Scenarios (see companion files)

File	User pwd	Owner pwd	Permissions	Use case
12a	user123	admin123	All	Protect authorship; allow all viewer operations (AES-256)
12b	(none)	ownerOnly	Accessibility	Distribute freely; prevent print / copy (AES-256)
12c	printme	printAdmin	Print only	Allow printing; prevent digital re-use (AES-256)
12d	viewonly	superadmin	None	Maximum restriction — view on screen only (AES-256)

TerraPDF — Encryption Showcase

Encryption & Password Protection

[Overview](#)

File	User pwd	Owner pwd	Permissions	Use case
12e	legacy123	legacyAdmin	All	Legacy AES-128 opt-in for pre-2008 viewers

PdfPermissions Flags

Combine flags with bitwise-OR. Use PdfPermissions.All to grant everything or PdfPermissions.None to deny all.

Flag	PDF bit	Description
PdfPermissions.Print	Bit 3	High-quality printing
PdfPermissions.ModifyContents	Bit 4	Modify document contents
PdfPermissions.CopyText	Bit 5	Copy or extract text and graphics
PdfPermissions.ModifyAnnotations	Bit 6	Add or modify annotations and form fields
PdfPermissions.FillForms	Bit 9	Fill in interactive form fields
PdfPermissions.ExtractForAccessibility	Bit 10	Text extraction for screen readers
PdfPermissions.AssembleDocument	Bit 11	Insert, rotate, or delete pages
PdfPermissions.PrintLowResolution	Bit 12	Low-resolution (degraded) printing only
PdfPermissions.All	Bits 3-12	All flags combined — full viewer access
PdfPermissions.None	Bits 3-12	No flags set — view on screen only

EncryptionOptions API

Property	Type	Default	Description
UserPassword	string?	null	Password to open the document. Omit for no open prompt.
OwnerPassword	string?	null	Full-access password. Auto-generated when null.
Permissions	PdfPermissions	All	Bitwise flags controlling viewer operations.

Complete API Example

```
Document.Create(container =>
{
    container.Encrypt(new EncryptionOptions
    {
        UserPassword = "open123",
        OwnerPassword = "admin456",
        Permissions = PdfPermissions.Print
        | PdfPermissions.CopyText,
    });

    container.Page(page =>
    {
        page.Size(PageSize.A4);
        page.Margin(2, Unit.Centimetre);
        page.Content().Text("Confidential").Bold();
    });
})
.PublishPdf("protected.pdf");
```